

Figure 11 — Causal Integration Ladder · Level II classification of TCGA-CESC priority genes

Per-gene synthesis of differential expression (TCGA-CESC) · GWAS Catalog tier · GTEx cis-eQTL · Open Targets colocalization (PP.H4) · Wald-ratio MR / published MR

Gene	TCGA DE	GWAS tier	cis-eQTL	Coloc · HPV/cervix	Coloc · cancer	Coloc · immune	MR evidence	CIL Level II
KRT5	▲ log2FC=+3.12	MODERATE	rs688861 NES=-0.26 p=3.6e-20	H4=0.98 Viral warts & HPV (PheCo...	H4=1.00 Basal cell carcinoma, in...	H4=1.00 Leukocyte-associated imm...	OR=1.017 p=8.3e-10	Full Level II
S100A8	▲ log2FC=+6.45	STRONG	rs3014874 NES=-0.07 p=1.2e-08	n.s.	n.s.	H4=1.00 monocyte (fraction, mean...	3 pub MR	Full Level II
S100A9	▲ log2FC=+6.99	STRONG	rs3014874 NES=-0.08 p=2.6e-13	n.s.	n.s.	H4=0.93 Monocyte count	1 pub MR	Full Level II
CDC20	▲ log2FC=+4.85	MODERATE	rs1199039 NES=-0.08 p=3.7e-06	n.s.	H4=0.95 Tumor necrosis factor re...	H4=0.97 White blood cell count	—	Partial II eQTL + coloc
CDKN2A	▲ log2FC=+4.60	MODERATE	rs3731198 NES=+0.21 p=2.8e-07	n.s.	H4=1.00 Acute lymphoblastic leuk...	H4=0.98 Lymphocyte count	—	Partial II eQTL + coloc
KRT6A	▲ log2FC=+4.08	MODERATE	rs11439938 NES=-0.45 p=4.3e-07	H4=0.98 Viral warts & HPV (PheCo...	H4=0.98 Basal cell carcinoma, in...	n.s.	OR=0.988 p=8.3e-10	Partial II eQTL + coloc
PITX1	▲ log2FC=+4.95	MODERATE	rs6596201 NES=+1.01 p=1.2e-21	n.s.	H4=0.99 Benign neoplasm of colon...	H4=0.95 Neutrophill percentage (...)	—	Partial II eQTL + coloc
SERPINB5	▲ log2FC=+4.20	LIMITED	rs6567350 NES=+0.15 p=5.0e-16	n.s.	n.s.	n.s.	—	Partial II eQTL/GWAS

Strong (PP.H4 ≥ 0.80 / p ≤ 5×10⁻⁸ / STRONG tier)

Moderate (PP.H4 0.70-0.80 / p ≤ 1×10⁻⁵ / MODERATE tier)

Limited (PP.H4 0.50-0.70 / suggestive / LIMITED tier)

Not significant / no evidence